Supplement to A Hierarchy of Risk Certificates for Forecasting Representations

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Abstract

This supplement preserves the complete secondary experimental exposition and the inherited local KAHM reconstruction equations accompanying the main manuscript. It includes all finite-support lower-bound comparisons, the explicit-feature decision control, full forecasting tables, preprocessing comparisons, matched geometries and decoders, and speech-energy evidence. Adverse and inconclusive outcomes are retained. All proofs of the main paper's stated results remain in the main PDF. The repository supplies frozen configurations, individual outcomes, exact source snapshots, and reproduction commands for each separately defined study.

Reading conventions. Sections, tables, figures, and equations in this supplement have an S prefix. References without that prefix refer to the main paper. Historical studies retain their original designs and confidence allocations. Repeated fitting on a previously examined natural-data cohort is descriptive; it does not create an independent confirmation sample. The source-generation step only expands the retained TeX inputs and resolves cross-document references.

S1 Local KAHM reconstruction implementation

This section records the inherited local reconstruction procedure used by the forecasting implementation. It is supplied for reproducibility; the manuscript's risk hierarchy requires the resulting fixed representation and metric, rather than this particular reconstruction algorithm. The notation follows the main paper: $z(x) \in \mathbb{R}^{d_x}$ is the training-preprocessed history, and $\widehat{z}_{bcj}(x)$ is the reconstruction from local model j in group c of retained dictionary block b.

For one such local model, suppress the indices (b, c, j) and write its $N_{\mathrm{loc}} \geq 2$ reference histories as $z_1, \ldots, z_{N_{\mathrm{loc}}} \in \mathbb{R}^{d_x}$, with at least one direction of nonzero empirical variance. The fitted linear encoding $P \in \mathbb{R}^{r \times d_x}$ maps histories to $r \geq 1$ principal-component coordinates. The encoding retains only principal-component directions with positive empirical variance. Let $\Sigma \in \mathbb{R}^{r \times r}$ be the resulting nonsingular sample covariance of the encoded references. The Gaussian kernel, its reference Gram matrix, and its query vector are

$$k(\xi, \xi') = \exp\left\{-\frac{(\xi - \xi')^\top \Sigma^{-1} (\xi - \xi')}{2r}\right\},$$
$$K_{i\ell} = k(Pz_i, Pz_\ell), \qquad k_x = (k(Pz_i, Pz(x)))_{i=1}^{N_{\mathrm{loc}}}.$$

Here $\xi, \xi' \in \mathbb{R}^r$. For the local regularization parameter $\eta > 0$, kernel regression gives $h(x) = (K + \eta I_{N_{\mathrm{loc}}})^{-1} k_x \in \mathbb{R}^{N_{\mathrm{loc}}}$. Its coordinate h_i uses regression targets equal to one at reference i and zero at the other references. The reconstruction and its normalizer are

$$s(x) = \sum_{i=1}^{N_{\mathrm{loc}}} h_i(x), \qquad \widetilde{s}(x) = \begin{cases} s(x), & s(x) \neq 0, \\ 1, & s(x) = 0, \end{cases}$$

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Law	Representation values u_i	Targets y_i	Stratum labels $c(u_i)$
Collapsed binary	$(0, 0)$	$(-1, 1)$	$(0, 0)$
Two atoms	$(-.6/\sqrt{2}, .6/\sqrt{2})$	$(-1, 1)$	$(0, 0)$
Three-point V	$(-1, 0, 1)$	$(1, -1, 1)$	$(0, 0, 1)$
Five-point zigzag	$(-1, -.5, 0, .5, 1)$	$(-1, 1, -1, 1, -1)$	$(0, 0, 1, 1, 1)$
Partial collapse	$(-1, -1, 0, 0, 1, 1)$	$(-1, .2, -.2, .2, -.2, 1)$	$(0, 0, 1, 1, 2, 2)$
Noisy affine	$(-1, -1, 0, 0, 1, 1)$	$(-1, 0, -.5, .5, 0, 1)$	$(0, 0, 1, 1, 2, 2)$
Rare conflict	$(3, 0, 1)$	$(0, -1, 1)$	$(0, 1, 1)$
Affine mismatch	$(-1, 0, 1)$	$(-.5, .5, -.5)$	$(0, 0, 0)$

Table S1: Prespecified joint supports for the lower-bound comparison. Entry i gives one possible pair $(U, Y) = (u_i, y_i)$. Each row assigns equal probabilities to its entries, except rare conflict, whose probabilities are $(.9, .05, .05)$. Repeated representation values retain different possible targets. Stratum labels depend only on U and are fixed before calibration. Every law uses $|u - u'|$, $\kappa = 1$, and predictions in $[-1, 1]$.

$$\widehat{z}(x) = \sum_{i=1}^{N_{\mathrm{loc}}} \frac{h_i(x)}{\widetilde{s}(x)} z_i.$$

The local regularization parameter also depends on the preprocessed values. Collect the reference histories in the matrix $Z = [z_1, \ldots, z_{N_{\mathrm{loc}}}]$ of size $d_x \times N_{\mathrm{loc}}$, and let $v_z = \|Z\|_{\mathrm{F}}^2/(d_x N_{\mathrm{loc}})$. Starting at $\eta_0 = 5v_z/2$, the implementation computes

$$B_k = (K + \eta_k I_{N_{\mathrm{loc}}})^{-1}, \qquad \eta_{k+1} = 2v_z + \frac{\|Z - ZKB_k\|_{\mathrm{F}}^2}{d_x N_{\mathrm{loc}}}.$$

It stops when $|\eta_{k+1} - \eta_k| \le 10^{-3}$ or after 100 iterations, and retains the last computed B_k; thus $\eta = \eta_k$ in the query map above. Nonzero empirical variation implies $v_z > 0$, so every inverse uses positive regularization. This parameter is separate from the forecast-decoder ridge parameter below. When $s(x) \neq 0$, the reconstruction coefficients sum to one and may be signed, so $\widehat{z}(x)$ is an affine combination of reference histories. The zero-sum convention defines the implementation at all queries but need not preserve this affine-combination property.

The main paper defines the radial and angular folding score and the stable normalized associations from these reconstructed histories. All local reference sets, encodings, covariance matrices, and regularizers are fitted on development data and frozen before calibration.

S2 Lower-bound comparisons beyond two atoms

The two-atom law in Proposition 3 has an exact population witness. To distinguish a loose population bound from finite-sample uncertainty, we evaluate eight finite-support distributions (Table S1). They include deterministic and noisy targets, histories assigned identical features despite different targets, rare conflicting targets, and a case in which affine decoders are less expressive than general Lipschitz decoders. For every method, R^* is the minimum MSE over the same full scalar 1-Lipschitz class with outputs in $[-1, 1]$. These explicit representations isolate the behavior of the certificates; they do not measure the forecasting accuracy of learned KAHM features.

	Population quantities				Mean lower endpoint, $n = 4096$			
Law	R^*	W	W_{str}	C	Pair	Centered	ERM	Strat.
Collapsed binary	1.0000	1.0000	1.0000	1.0000	0.9306	0.9149	0.9345	0.9237
Two atoms	0.3315	0.3315	0.3315	0.3315	0.3034	0.2807	0.2908	0.3001
Three-point V	0.2222	0.2222	0.1667	0.0160	0.1996	0.0000	0.1890	0.1355
Five-point zigzag	0.5400	0.3800	0.5250	0.0744	0.3408	0.0186	0.4891	0.4442
Partial collapse	0.2533	0.0867	0.2533	0.0000	0.0646	0.0000	0.2178	0.1772
Noisy affine	0.2500	0.0972	0.2500	0.0000	0.0777	0.0000	0.2145	0.1801
Rare conflict	0.0250	0.0025	0.0250	0.0000	0.0000	0.0000	0.0148	0.0059
Affine mismatch	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table S2: Finite-support comparison on the same full scalar Lipschitz class. R^* is the minimum MSE, W the unconditional pair witness, W_{str} the prespecified stratified witness, and C the centered population bound. The last four columns average 300 independent calibration samples. Each method separately has nominal one-sided confidence 95%; no joint confidence across columns is asserted. Empirical risk minimization (ERM) uses a dual-certified empirical minimum and bounded-loss Kullback–Leibler (KL) inversion. Primary stratification estimates the stratum masses. All quantities are in MSE units.

Protocol and comparators. The protocol was recorded before sampling. Each law uses 300 independent calibration samples at each $n \in \{64, 256, 1024, 4096\}$, totaling 9,600 trials. The four primary methods share each sample and each receive a separate one-sided failure probability 0.05; no joint 95% event across methods is asserted. We compare the disjoint-pair empirical Bernstein (EB) bound in Theorem 7, the range-aware centered bound in Appendix B.2, the dual-certified empirical risk minimization (ERM) bound with Kullback–Leibler (KL) correction in (39), and the stratified certificate in Theorem 8. The primary stratified method estimates the stratum probabilities from the calibration data. Both primary pairwise methods use the generic score range $[0, 2]$, obtained from a target diameter of two. Prespecified secondary analyses tighten this range using the known support, use known stratum masses, or replace the empirical-minimum correction by Hoeffding's bound. The result archive includes every secondary analysis.

The empirical and population optimization programs use the full Lipschitz class, with global extension from the finite support. Feasible primal and dual values bracket each optimum; the largest empirical gap is 4.01×10^{-11}. Analytic checks cover all eight population optima. An unprojected affine optimum is never used as a Lipschitz lower bound: in the mismatch law, the former is $2/9$ and the latter is zero.

Population slack and finite-sample cost. Table S2 reports population quantities separately from mean calibrated endpoints. On partial collapse, stratification raises the population witness from 0.0867 to the exact minimum 0.2533. At $n = 4096$, its mean endpoint is 0.1772, compared with 0.0646 for unconditional pairing and 0.2178 for empirical minimization. Stratification also improves the five-point and noisy-affine witnesses. It worsens the three-point witness, from 0.2222 to 0.1667, demonstrating that a fixed partition need not help.

Empirical minimization gives the largest mean endpoint on most richer laws; the pair bound is larger on the partially separated two-atom and three-point laws at $n = 4096$. Accounting for a

known representation range substantially improves centered calibration when the representation is constant. The two-atom comparison in Section S4 quantifies how treating a known zero variance as unknown can weaken a centered endpoint. Consequently, the relative performance of pairwise and centered certificates depends on the estimator and the structural information supplied to it.

Positive tolerances, coverage, and computation. The fixed RMSE tolerances are 0.10, 0.25, and 0.50 for every law. On partial collapse at $n = 4096$, unconditional pairing excludes tolerance 0.25 in 211 of 300 trials; empirical minimization and primary stratification both exclude it in all 300. Rare conflicts remain difficult: their exact risk is 0.025, but unconditional pairing and primary stratification never exclude tolerance 0.10 at this sample size. Empirical minimization excludes it in all 300 trials. Tightening the stratified score range using the known support raises its exclusion count to 284; additionally supplying known masses raises it to 300. The gain therefore depends on both pairing and available information.

All four primary methods have zero observed coverage violations across the prespecified grid. This simulation does not prove coverage; each law–size cell has only 300 repetitions, and the result archive gives binomial Monte Carlo confidence intervals. At $n = 4096$, medians across the eight law-wise median computation times are 31.3, 34.7, 265.4, and 165.0 μs for pair EB, centered, empirical minimum, and stratified EB, respectively. These single-machine measurements use a precomputed finite-support pair-score table and include sample statistics, confidence corrections, and ERM optimization after representation lookup. They exclude feature learning and do not establish a computational advantage on large continuous supports. The frozen protocol, sufficient statistics, dual certificates, and complete analysis are in `results/witness_comparison`.

S3 Decisions at a positive error tolerance

Decisions in the KAHM forecasting studies. The main synthetic, oscillator, and coefficient-budget studies contain 62 fitting runs (Sections S4–S4.4). We distinguish exclusion of a candidate's decoder class from the behavior of a model selected by validation. Among the 800 main-study candidates, the all-horizon decisions at RMSE tolerance 0.35 are 301 accepted, 24 excluded, and 475 unresolved; the corresponding oscillator counts are 55, 30, and 155 among 240 candidates. Nevertheless, all 186 horizon-specific lower endpoints of the 62 models selected by validation across the complete candidate family are zero. Restricting validation selection to retained KAHM gives 184 zero endpoints out of 186. These descriptive counts motivate a separate test of whether lower certificates can resolve a decision that upper calibration leaves open.

A controlled forecasting mechanism. The protocol is specified before calibration observations are generated. Each independent episode supplies one history $X = (Z, S)$ and one scalar future Y. The indicator S equals one with probability 0.2 and zero otherwise, and Z, R, N are mutually independent random signs, each taking -1 and 1 with equal probability, independently of S. We consider two regimes, called repairable and irreducible because the final representation can remove the signal uncertainty only in the first:

$$Y = 0.8SZ + 0.1N \quad \text{(repairable)}, \qquad Y = 0.8SR + 0.1N \quad \text{(irreducible)}.$$

The three entries of (Z, S, Y) occur in temporal order, and the forecast is made after observing Z and S, before observing Y. The target range is $[-0.9, 0.9]$. We append the simplex blocks

$$(1), \qquad (1 - S, S), \qquad \big((1 + SZ)/2, (1 - SZ)/2\big)$$

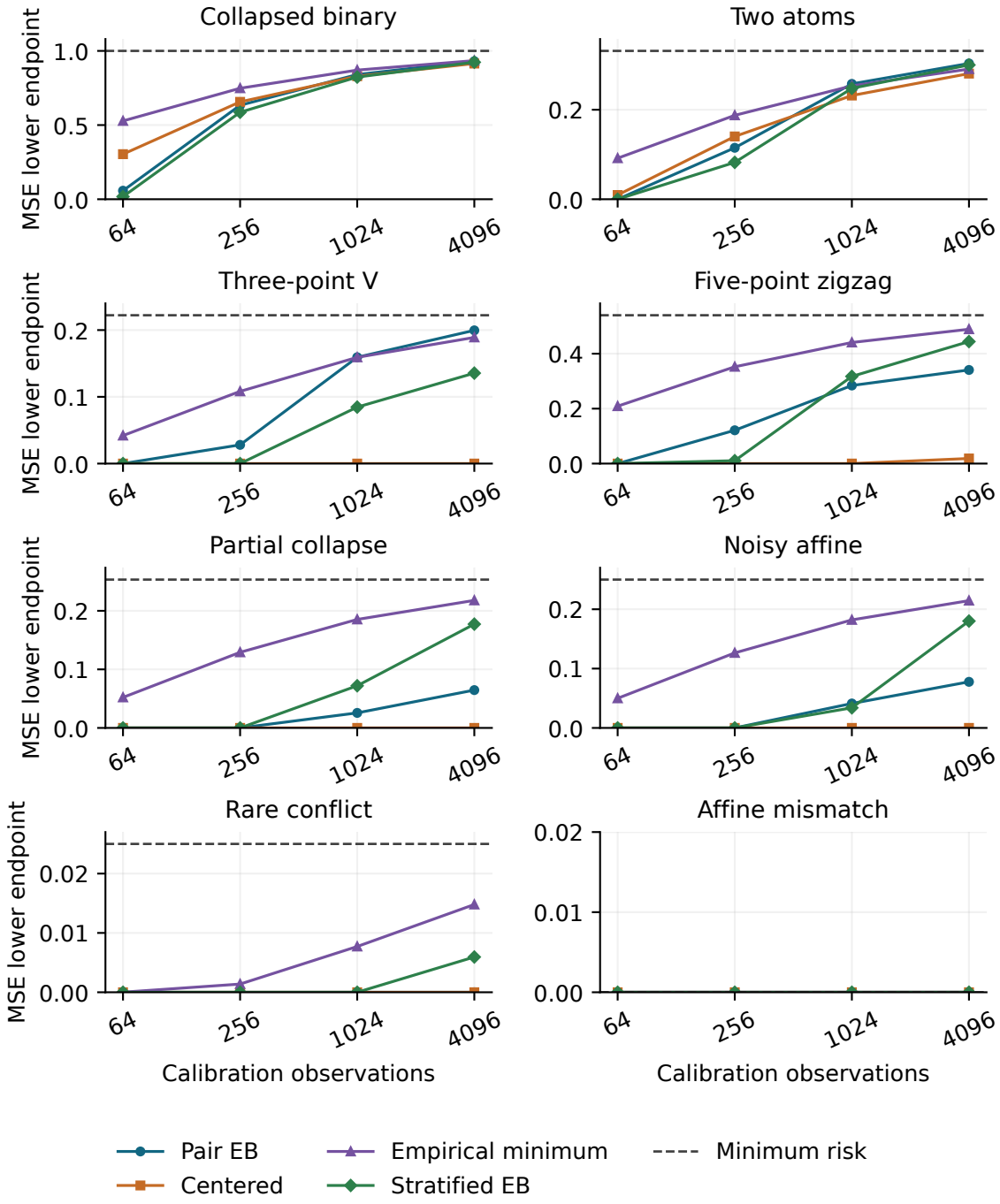


Figure S1: Mean lower endpoints over all 300 repetitions at each calibration size for every prespecified law. Dashed lines show minimum population MSE. Each method has its own 95% one-sided confidence statement. The result archive provides endpoint quantiles, Monte Carlo confidence intervals, secondary methods, and all individual trials.

n	Repairable regime			Irreducible regime	
	Upper acceptance	Minimum cost 5		All classes excluded	
		Unconditional	Stratified	Unconditional	Stratified
256	0	0	0	0	0
1,024	100	0	0	0	0
4,096	100	0	0	0	0
16,384	100	0	100	0	1
65,536	100	0	100	0	100

Table S3: Decision counts out of 100 independent repetitions in each constructed regime, at the prespecified RMSE tolerance 0.30. Upper acceptance certifies a fitted predictor; equality of the cost endpoints certifies the least feasible cost among the three predefined classes. The unconditional and stratified methods share the same upper endpoint and fitted model. Every irreducible-regime upper decision is unresolved. One failure budget 0.05 covers all five calibration sizes and all reported endpoints within each regime and repetition.

under the common coefficient budget $\kappa = 2$. The first block has one coordinate; the first two have three coordinates in total; and all three have five. These coordinate counts define the costs $1, 3, 5$ of the three representation classes. These explicit representations test the general theorem; they are not fitted KAHM dictionaries.

At costs 1, 3, and 5, the exact minimum MSEs in the repairable regime are 0.138, 0.138, and 0.01. They all equal 0.138 in the irreducible regime. In the former, the last block represents the conditional mean $0.8SZ$ with coefficients $(0.8, -0.8)$, whose norm is below 2. In the latter, the unobserved sign R leaves the conditional mean zero even with the complete history. The prescribed RMSE tolerance 0.30 therefore requires the last prefix in one regime and is unattainable in the other. For the cost-3 prefix, the unconditional population witness is only 0.0324, whereas pairing conditionally on S gives 0.138. This isolates a population limitation of unconditional pairing, in addition to finite-sample uncertainty.

Calibration and comparisons. Each of 100 independent repetitions per regime uses 2,048 training episodes, 1,024 validation episodes, and nested calibration samples of $256, 1024, 4096, 16384, 65536$ episodes. All three decoders are fitted with zero ridge penalty and frozen before calibration. The RMSE tolerance 0.30 corresponds to an MSE threshold of 0.09. We compare validation-only acceptance at this threshold, calibrated upper-bound acceptance, and two extensions of the latter rule using either unconditional or stratified lower certificates. Validation-only acceptance is a descriptive reference rule without a population confidence guarantee. The two predefined partitions use the history labels S and (S, Z); neither partition uses the future target. For the latter method, we estimate stratum masses and take the maximum of the two separately budgeted endpoints from Theorem 8. One failure budget 0.05 per regime and repetition covers all prefixes and all five sample sizes: 0.02 for upper bounds, 0.01 for unconditional lower bounds, and 0.02 for both stratified families. Calibration uses one forecast per independent episode; it does not treat dependent windows as independent observations.

Decision value and sample requirements. Validation accepts the repaired cost-5 model in all 100 repetitions, and upper calibration does so from $n = 1024$ onward. Stratified lower certificates exclude

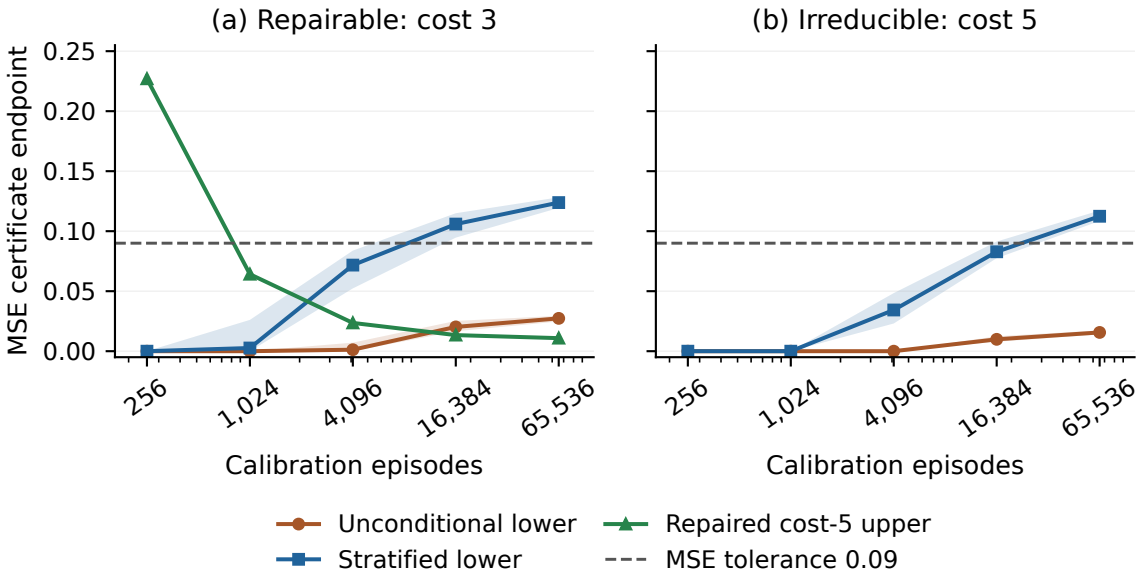


Figure S2: Calibration-size dependence in the constructed decision experiment. Lines show medians and shaded bands the full range across 100 independent repetitions; the bands describe repetition variability. Panel (a) compares lower endpoints for the inadequate cost-3 class with the upper endpoint for the repaired cost-5 predictor. Panel (b) shows lower endpoints for the most detailed class in the irreducible regime. The dashed line is the fixed MSE tolerance 0.09.

both cheaper classes in all repetitions at $n = 16384$ and 65536, so the cost interval in Corollary 11 reduces to $[5, 5]$. Unconditional lower calibration never closes this interval at the prescribed sample sizes. Both procedures select the same accepted predictor; the additional conclusion is that no cheaper prescribed class can meet the tolerance, independently of its decoder optimization.

The irreducible control produces no upper acceptance. Stratified lower calibration excludes all three classes in 1 of 100 repetitions at $n = 16384$ and all 100 at $n = 65536$; unconditional lower calibration never excludes all three. Small samples reveal the cost of stratification: at $n = 4096$, unconditional pairing excludes the constant class in every repetition of both regimes, while the stratified method excludes no class. All 600 fits converged, and the exact population risks satisfy every reported simultaneous interval event in all 200 regime–repetition instances. The experiment demonstrates a class-level decision under a controlled mechanism, with substantial sample requirements; it does not establish its frequency in natural forecasting tasks.

S4 KAHM forecasting on bounded synthetic systems

The preceding studies test the general risk certificates. This section applies the KAHM learner to four bounded forecasting systems. The comparison includes all prescribed configurations. Empirical test error describes performance on a held-out sample; a certified risk endpoint bounds the corresponding population expectation under the stated sampling assumptions.

Method	Harmonic		Quasiperiodic		Nonlinear AR		Logistic	
	$h = 1$	$h = 12$	$h = 1$	$h = 12$	$h = 1$	$h = 12$	$h = 1$	$h = 12$
KAHM blocks	0.070 (0.001)	0.346 (0.006)	0.078 (0.003)	0.284 (0.004)	0.054 (0.001)	0.074 (0.001)	0.176 (0.018)	0.511 (0.001)
RBF blocks	0.068 (0.001)	0.351 (0.003)	0.068 (0.005)	0.287 (0.004)	0.057 (0.001)	0.074 (0.001)	0.172 (0.012)	0.512 (0.001)
Affine-hull blocks	0.072 (0.003)	0.357 (0.004)	0.110 (0.007)	0.275 (0.006)	0.059 (0.003)	0.074 (0.001)	0.156 (0.014)	0.509 (0.001)
Hard KAHM states	0.097 (0.003)	0.358 (0.003)	0.127 (0.002)	0.300 (0.006)	0.062 (0.001)	0.074 (0.001)	0.235 (0.005)	0.512 (0.001)
Ridge AR	0.038 (0.000)	0.281 (0.004)	0.033 (0.000)	0.259 (0.002)	0.047 (0.000)	0.080 (0.001)	0.409 (0.002)	0.507 (0.002)
Quadratic regression	0.031 (0.000)	0.243 (0.003)	0.030 (0.000)	0.246 (0.002)	0.047 (0.000)	0.081 (0.001)	0.010 (0.000)	0.508 (0.003)
KAHM rollout	0.070 (0.002)	0.358 (0.004)	0.079 (0.004)	0.319 (0.011)	0.055 (0.001)	0.074 (0.001)	0.174 (0.018)	0.524 (0.004)
Polynomial rollout	0.030 (0.000)	0.297 (0.010)	0.030 (0.000)	0.288 (0.004)	0.047 (0.000)	0.084 (0.003)	0.010 (0.000)	0.502 (0.004)
Persistence	0.125 (0.000)	1.085 (0.004)	0.084 (0.000)	0.743 (0.004)	0.055 (0.000)	0.103 (0.001)	0.931 (0.006)	0.661 (0.006)

Table S4: Independent-test RMSE: mean over five complete fitting repetitions, with sample standard deviation below in parentheses. One candidate per family is selected by validation MSE across all three horizons before calibration or test scoring. The archive includes $h = 4$, all candidates, and every confidence endpoint. Fitting variation differs from uncertainty conditional on a fitted model; these test errors are descriptive estimates.

S4.1 Protocol and predictive accuracy

Four generators produce independent length-72 episodes: a noisy harmonic oscillator, a quasiperiodic signal, nonlinear autoregression with bounded innovations, and a logistic map with an episode-specific parameter. Their observations lie in $[-1, 1]$ by construction (Appendix S8). Each of five fitting repetitions uses 128 training, 512 validation, 4096 calibration, and 2048 test episodes. Histories have lengths $m \in \{4, 12\}$, horizons are $\{1, 4, 12\}$, and the 13 common zero-based origins are $11, 15, \ldots, 59$. Within-episode windows may be dependent. Geometry is relearned in each repetition, with budget $\kappa = 4$, ridge coefficient 10^{-4}, affinity exponent 16, and prefix dimensions $2, 6, 14, 30$. Geometry fitting uses at most 1600 histories and 24 local points.

Each repetition freezes 40 candidates: retained KAHM, RBF, and affine-hull features; hard KAHM assignments; ordinary Frobenius-budget and exact-duplicate ablations; ridge autoregression (AR); quadratic regression; KAHM and polynomial feature rollouts; persistence; and a constant model. One candidate per family is selected by validation MSE across horizons. These specified configurations assess the procedure, rather than the best possible tuning of each family.

The failure budget 0.05 per repetition allocates 0.04 across $2 \times 40 \times 3$ one-sided risk bounds and 0.01 across 18 adjacent-prefix improvement comparisons. Upper bounds use 4096 episode losses; lower bounds use 2048 disjoint episode pairs. The 13 origins do not increase these independent sample sizes. The joint event is per complete repetition, not across all runs.

	KAHM retained family			All candidates		
System	$\epsilon = .20$	.35	.50	$\epsilon = .20$	.35	.50
Harmonic	0/5 (–)	0/5 (–)	5/5 (6)	0/5 (–)	5/5 (12)	5/5 (4)
Quasiperiodic	0/5 (–)	5/5 (14)	5/5 (2)	0/5 (–)	5/5 (4)	5/5 (1)
Nonlinear AR	5/5 (2)	5/5 (2)	5/5 (2)	5/5 (2)	5/5 (2)	5/5 (2)
Logistic	0/5 (–)	0/5 (–)	0/5 (–)	0/5 (–)	0/5 (–)	0/5 (–)

Table S5: Runs certifying the same RMSE tolerance at all horizons $1, 4, 12$. Parentheses give median retained feature dimension among successful least-dimension selections; '–' means no model passed. Cost is a representation-size proxy, not measured runtime. Each run has a joint 95% event covering its complete frozen family and improvement comparisons.

Table S4 includes variation across the five fits. KAHM improves substantially over long-horizon persistence on the periodic generators, but quadratic and autoregressive models are stronger there. On nonlinear AR, KAHM horizon-12 RMSE is 0.0743, compared with 0.0802 for ridge AR and 0.0813 for quadratic regression; the latter families are better at horizon one. Direct KAHM fitting improves on feature rollout at horizon 12 on the two periodic generators. The independently initialized oscillators in Section S4.3 show that direct fitting does not always outperform feature rollout.

S4.2 Acceptance, exclusion, and fitted refinement

The retained KAHM family certifies the all-horizon RMSE requirement 0.20 for every nonlinear-AR repetition, 0.35 for every quasiperiodic repetition, and 0.50 for every harmonic repetition. Allowing all families also certifies 0.35 on harmonic data. No candidate certifies 0.50 on the logistic task; this is unresolved, not evidence that all true risks exceed the threshold. Even the selected harmonic quadratic model's mean one-step test RMSE 0.0311 corresponds to a mean upper endpoint 0.1497, illustrating the precision cost.

Of 480 retained-KAHM candidate–horizon statements, 119 have positive lower endpoints. Only two exclude RMSE 0.35, and none excludes 0.50; positive endpoints alone are not enough for a useful positive-tolerance decision. Section S3 compares the acceptance, exclusion, and unresolved decisions across the forecasting studies. Paired comparisons certify 72 of 360 adjacent-prefix improvements. Nonlinear AR has no certified adjacent-prefix gain despite meeting useful tolerances. The complete refinement curves and system-specific counts are retained in the repository; nesting a class does not guarantee improvement of each fit.

Exact duplication is already covered analytically by Theorem 12. Zero-ridge numerical checks reproduce matching minimum training risks within their solver gaps. The main duplicate candidate uses zero ridge, whereas primary models use 10^{-4}; their fitted errors are therefore not evidence for ridge-estimator invariance. All 800 main-study candidates completed; the constrained block fits have recorded first-order optimality gaps below 10^{-7}. These numerical optimization guarantees are separate from the risk guarantees, which require the sampling assumptions in Section 3. Appendix S8 describes reproduction of the fitted models and confidence endpoints.

Calibration on a two-atom distribution. A separate benchmark contains 28,800 trials with analytically known class and fitted risks (Appendix D). One pairwise lower endpoint exceeds its true risk; no centered lower or fitted upper endpoint does. The centered estimator in this benchmark

System	Family	$h = 1$	$h = 4$	$h = 12$	$\overline{\sqrt{U_{12}}}$
Van der Pol	KAHM blocks	0.099 ± 0.001	0.139 ± 0.005	0.381 ± 0.023	0.409
Van der Pol	RBF blocks	0.106 ± 0.005	0.152 ± 0.011	0.383 ± 0.010	0.412
Van der Pol	Affine blocks	0.137 ± 0.003	0.192 ± 0.006	0.299 ± 0.009	0.333
Van der Pol	Ridge AR	0.038 ± 0.000	0.150 ± 0.001	0.353 ± 0.001	0.383
Van der Pol	Quadratic direct	0.032 ± 0.000	0.138 ± 0.001	0.342 ± 0.001	0.374
Van der Pol	KAHM rollout	0.034 ± 0.012	0.064 ± 0.030	0.341 ± 0.033	0.374
Van der Pol	Quadratic rollout	0.029 ± 0.000	0.127 ± 0.002	0.399 ± 0.006	0.427
Van der Pol	Persistence	0.089 ± 0.000	0.339 ± 0.000	0.868 ± 0.001	0.885
Duffing	KAHM blocks	0.080 ± 0.002	0.120 ± 0.004	0.250 ± 0.010	0.300
Duffing	RBF blocks	0.061 ± 0.001	0.098 ± 0.003	0.229 ± 0.007	0.281
Duffing	Affine blocks	0.064 ± 0.004	0.097 ± 0.003	0.223 ± 0.005	0.274
Duffing	Ridge AR	0.019 ± 0.000	0.041 ± 0.000	0.151 ± 0.001	0.214
Duffing	Quadratic direct	0.017 ± 0.000	0.035 ± 0.000	0.137 ± 0.001	0.203
Duffing	KAHM rollout	0.081 ± 0.001	0.126 ± 0.004	0.269 ± 0.014	0.315
Duffing	Quadratic rollout	0.017 ± 0.000	0.035 ± 0.000	0.140 ± 0.002	0.206
Duffing	Persistence	0.033 ± 0.001	0.114 ± 0.002	0.318 ± 0.005	0.363

Table S6: Independent-test sensor RMSE, mean $\pm$ sample standard deviation over three complete fits, with validation selection within each family. The last column averages the selected models' individual horizon-12 upper-RMSE endpoints; this descriptive average gives no ensemble guarantee. All 40 candidates per repetition are archived.

estimates representation variance even when that variance is known to be zero. At full collapse and $n = 4096$, its mean lower endpoint is 0.5327. Substituting the known zero variance while keeping the same target correction gives a lower bound of approximately 0.9031 on the expected endpoint; the pairwise method has mean endpoint 0.9177. Thus part of the observed difference is caused by unused structural information. The $n^{-1/4}$ correction of this centered estimator is not a limitation of all centered methods. Section S2 compares a method based on direct standard-deviation concentration and a certified empirical minimum.

S4.3 Independently initialized Van der Pol and Duffing episodes

We apply the frozen 40-candidate procedure to independently initialized Van der Pol and forced Duffing episodes, connecting the experiment to Kumar and Kargaran (2026). Each system has three new complete fits with the sample sizes and confidence allocation of Section S4. The bounded position sensor, dynamics, and numerical checks are specified in Appendix S8.2. Horizons $1, 4, 12$ correspond to $0.1, 0.4, 1.2$ time units. Certificates concern sensor-output error under the independent-episode numerical law; they require neither stationarity nor independent within-episode windows and do not certify physical-state error.

Retained KAHM improves horizon-12 RMSE over persistence on Van der Pol (0.3814 versus 0.8677) and Duffing (0.2499 versus 0.3185), but other models perform better: affine-hull blocks achieve 0.2990 on Van der Pol, and quadratic regression achieves 0.1369 on Duffing (Table S6). Iterated KAHM achieves 0.3408 on Van der Pol and 0.2686 on Duffing, so direct fitting outperforms iteration only on the latter system.

System	κ	Features	$h = 1$	$h = 4$	$h = 12$	Passes
Harmonic	1	16.7	0.3737	0.4102	0.4696	0/3
Harmonic	4	30.0	0.1414	0.2313	0.3900	0/3
Harmonic	16	30.0	0.1392	0.2292	0.3875	0/3
Quasiperiodic	1	24.7	0.2510	0.2905	0.3336	0/3
Quasiperiodic	4	30.0	0.1294	0.2050	0.2927	3/3
Quasiperiodic	16	30.0	0.1294	0.2050	0.2927	3/3
Nonlinear AR	1	7.3	0.1059	0.1056	0.1068	3/3
Nonlinear AR	4	30.0	0.0670	0.0730	0.0746	3/3
Nonlinear AR	16	30.0	0.0670	0.0730	0.0746	3/3
Logistic	1	30.0	0.4713	0.5003	0.5052	0/3
Logistic	4	30.0	0.4384	0.4967	0.5044	0/3
Logistic	16	30.0	0.4384	0.4968	0.5044	0/3

Table S7: Budget sensitivity: mean test RMSE and feature count over three new fitting seeds. "Passes" counts validation-selected KAHM models certified at tolerance 0.35 simultaneously for $h = 1, 4, 12$. The confidence allocation covers all three budgets within each system–seed repetition.

Retained KAHM certifies all-horizon RMSE 0.35 in all three Duffing fits and 0.50, but not 0.35, in all three Van der Pol fits. Allowing all comparison families certifies 0.35 in every fit of both systems; none certifies 0.20. Adjacent KAHM refinements have positive certified improvement in 43 of 54 Van der Pol comparisons and 10 of 54 Duffing comparisons.

S4.4 Sensitivity to the coefficient budget

The budget κ defines a feature-dependent class; equal numerical budgets do not imply equal capacity across representations. To assess sensitivity to this choice, a separate experiment uses fresh development, calibration, and test streams with seeds 2701–2703, $m = 12$, and $\kappa \in \{1, 4, 16\}$ on all four main systems. Other settings match Section S4. The design comprises 36 runs and 756 fitted models, with 21 candidates per budget and system–seed combination. Budgets share streams and geometry within each system–seed repetition. Validation selects a KAHM prefix separately at each budget. Allocating $0.05/3$ to each budget's risk and refinement bounds gives joint confidence at least 0.95 across budgets within a repetition, without a joint claim across all twelve repetitions.

Increasing κ from 1 to 4 reduces one-step KAHM RMSE from 0.3737 to 0.1414 on the harmonic system and from 0.1059 to 0.0670 on nonlinear autoregression; quasiperiodic 0.35 certification increases from zero to three passes. Increasing it from 4 to 16 gives little further improvement: harmonic horizon-12 RMSE changes from 0.3900 to 0.3875, with no 0.35 pass; quasiperiodic and nonlinear-AR errors are unchanged at displayed precision, and logistic horizon-12 RMSE remains about 0.5044, without a pass. This does not establish impossibility at larger budgets or richer geometries.

At $\kappa = 16$, harmonic AR and quadratic baselines achieve horizon-12 RMSEs 0.2377 and 0.2375 and certify 0.35 in all three fits. On nonlinear autoregression at $\kappa = 4$, KAHM has lower horizon-12 RMSE (0.0746 versus 0.0800 and 0.0809) but higher one-step error. Every constrained fit has a first-order optimality gap below 10^{-7}.

Family	Features	$h = 1$	$h = 2$	$h = 4$
KAHM blocks	30.0	0.0466 ± 0.0010	0.0669 ± 0.0013	0.1127 ± 0.0013
RBF blocks	30.0	0.0421 ± 0.0017	0.0626 ± 0.0019	0.1105 ± 0.0015
Affine-hull blocks	30.0	0.0510 ± 0.0152	0.0664 ± 0.0114	0.1039 ± 0.0053
KAHM, Frobenius budget	30.0	0.0466 ± 0.0010	0.0669 ± 0.0013	0.1127 ± 0.0013
KAHM duplicate blocks	8.0	0.1386 ± 0.0016	0.1532 ± 0.0018	0.1822 ± 0.0018
KAHM hard assignment	16.0	0.0641 ± 0.0004	0.0856 ± 0.0009	0.1297 ± 0.0014
KAHM Koopman rollout	30.0	0.0473 ± 0.0011	0.0688 ± 0.0018	0.1177 ± 0.0025
Ridge autoregression	8.0	0.0130 ± 0.0000	0.0273 ± 0.0000	0.0679 ± 0.0001
Quadratic regression	14.0	0.0077 ± 0.0000	0.0178 ± 0.0001	0.0518 ± 0.0001
Quadratic Koopman rollout	14.0	0.0077 ± 0.0000	0.0185 ± 0.0002	0.0551 ± 0.0005
Persistence	4.0	0.0351 ± 0.0000	0.0696 ± 0.0000	0.1347 ± 0.0000
Constant	1.0	0.2207 ± 0.0007	0.2231 ± 0.0007	0.2273 ± 0.0008

Table S8: Original PenDigits forecasting on 1,751 eligible digits from 13 held-out writers. Coordinate-pooled test RMSE is mean $\pm$ sample standard deviation over three fitting seeds. "Features" is the mean retained count for validation-selected models. Each scalar forecast uses only its own coordinate's history; all selections precede test scoring.

S4.5 Writer-disjoint forecasting of original pen trajectories

We use the original, unnormalized PenDigits trajectories (Alpaydin and Alimoglu, 1996); the archive's spatially resampled table uses whole-character preprocessing unsuitable for causal forecasting. The known coordinate range $[0, 500]^2$ permits transforming each coordinate z to $z/250 - 1$ and clipping predictions to $[-1, 1]$, without stroke-specific normalization. Acquisition and development details are in Appendix S8.3.

Writer identifiers are inferred from the second integer in each UNIPEN `COMMENT` field: their ranges and record counts agree with the documented 30 development and 14 test writers, but the field semantics are undocumented. We retain the first 32 points of a first pen-down stroke only if its completed length is at least 32. This leaves 1,751 test digits from 13 eligible writers; one of the 14 test writers has no qualifying stroke. The task is retrospective and restricted to writers with eligible strokes; it does not cover every newly started stroke.

Each scalar forecast uses its own coordinate history at horizons $1, 2, 4$. For each of the two coordinates and three fitting seeds, the procedure fits 40 candidates and selects one per family by validation. All candidates and selection rules are frozen before testing. Losses average over origins and digits within writers, then equally over writers and coordinates before taking the square root. Table S8 reports the mean and sample standard deviation over three fitting seeds on the same test cohort; the deviations measure fitting variability, not variation across independent writer cohorts.

Quadratic regression is most accurate at every horizon: its mean RMSEs are 0.00768, 0.01777, and 0.05184, compared with 0.04661, 0.06686, and 0.11269 for retained KAHM. Ridge AR also outperforms KAHM. Compared with persistence, KAHM is worse at one step and better at two and four steps. Retained-block and ordinary Frobenius errors are effectively identical; selected block-norm sums below $\kappa = 4$ are consistent with an inactive constraint.

Population bounds additionally assume eligible writer records are iid from a common population and independent of development; the documentation does not establish this assumption. Writer-

averaged scalar losses lie in $[0, 4]$. The prescribed 0.05 allocation over 720 one-sided statements adds 2.42741 to each empirical MSE, clipping upper endpoints at 4 (Appendix S8.3). Pooled upper-RMSE endpoints near 1.56 certify none of the tolerances $0.2, 0.35, 0.5$. Neither the 1,751 trajectories nor their overlapping windows increase the independent-writer count. The experiment therefore supports descriptive writer-disjoint comparisons and illustrates limited certification with few groups. The fitted coefficient matrices satisfy their budgets, with first-order optimality gaps below 10^{-7}.

S5 Input scaling as a model choice

Training-based transformations. Section 5 explains why the complete KAHM representation is sensitive to input scaling. The forecasting experiments in Section S4 standardize each history coordinate using its training mean and standard deviation. We compare this with training min–max scaling, its clipped version, and raw histories. Here "raw" means no additional history transformation after the fixed observation transformation defining the forecasting task. For a history vector x and coordinate index j, let a_j and b_j be that coordinate's minimum and maximum over all training histories. The primary alternative is

$$z_j(x) = \begin{cases} 2(x_j - a_j)/(b_j - a_j) - 1, & b_j > a_j, \\ 0, & b_j = a_j. \end{cases} \tag{S1}$$

Each nonconstant training coordinate therefore spans $[-1, 1]$. A coordinate that is constant in training is mapped to zero for every query. Extrema are fixed before geometry subsampling and never recomputed on validation, calibration, or test trajectories. Unseen values may lie outside $[-1, 1]$; the diagnostic clipped version uses $\max\{-1, \min\{1, z_j(x)\}\}$. Clipping can map distinct out-of-range histories to the same input and is therefore a different representation, not merely a numerical safeguard. Only KAHM input geometry and models using it change in this comparison. Targets, budgets, data streams, and non-KAHM controls remain fixed. History and prefix selection use validation MSE within each preprocessing arm; test errors do not select an arm.

The first comparison reuses every original fitting seed and budget condition. Because this comparison was designed after inspecting the corresponding results, its calibration endpoints do not carry a confirmatory confidence guarantee for the subsequent choice of preprocessing. A fixed confirmation protocol uses new seeds 4701–4703 on all six synthetic systems, with fresh development, calibration, and test episodes. PenDigits instead uses new training subsamples on the same writer cohort; those comparisons remain descriptive.

Results. Table S9 reports all tasks. Mean MSE decreases on harmonic, quasiperiodic, and Duffing data in both stages; Duffing reductions are 7.07% and 7.29%. Logistic MSE increases by 2.65% and 6.97%, and Van der Pol MSE increases by 3.82% and 2.26%. The nonlinear-AR effect changes sign. Mean changes can disagree with a majority of seed-specific changes, so every paired effect and the sample standard deviation across fits are archived alongside conditional episode standard errors. Large numbers of test episodes do not replace independent fitting repetitions.

PenDigits soft-KAHM MSE decreases by 19.79% and 19.65%. Each coordinate improves at every horizon in each fit after averaging over writers. All 13 writer means improve after averaging coordinates and horizons, although horizon-four losses worsen in three original and two new-seed writer–fit comparisons. Ridge AR and quadratic regression remain more accurate. Hard KAHM

	Original fitting seeds				New fitting seeds			
Task	Standard	Min–max	$\Delta\%$	Wins	Standard	Min–max	$\Delta\%$	Wins
Harmonic	0.049148	0.048150	-2.03	4/5	0.050240	0.049586	-1.30	3/3
Quasiperiodic	0.037040	0.036182	-2.32	3/5	0.038290	0.035842	-6.39	3/3
Nonlinear AR	0.004561	0.004569	+0.18	4/5	0.004718	0.004656	-1.31	2/3
Logistic	0.164577	0.168931	+2.65	1/5	0.164632	0.176103	+6.97	0/3
Van der Pol	0.058343	0.060572	+3.82	0/3	0.054809	0.056049	+2.26	1/3
Duffing	0.027776	0.025812	-7.07	3/3	0.027913	0.025877	-7.29	3/3
PenDigits	0.006449	0.005173	-19.79	3/3	0.006792	0.005457	-19.65	3/3

Table S9: Validation-selected retained KAHM predictors: mean test MSE over all three horizons and fitting seeds. The primary alternative is training min–max scaling without query clipping. $\Delta\% = 100(\mathrm{MSE}_{\mathrm{minmax}}/\mathrm{MSE}_{\mathrm{standard}} - 1)$; negative values denote improvement. Wins counts fitting repetitions with a smaller all-horizon MSE. Original main tasks have five repetitions; original oscillators and PenDigits have three. All new-seed comparisons have three. Synthetic confirmation uses new development, calibration, and test episodes. PenDigits confirmation uses new training subsamples and the same held-out writer cohort, with coordinate MSE pooled before averaging horizons. Original-seed results reuse previously evaluated streams and are descriptive. No average across different task units is taken.

assignments instead worsen by 3.96% and 4.14% in every fit. Rollout improves on 7 of seven original tasks and 6 of seven new-seed tasks, but the new-seed logistic rollout worsens by 3.61%.

Clipped and unclipped min–max results are similar; raw inputs retain the logistic and Van der Pol limitations. Budget-specific effects also vary. Neither preprocessing nor budget is selected by test performance. All horizon-specific, uncertainty, diagnostic-arm, and budget tables remain in the reproducible result archive. At tolerances $0.20, 0.35, 0.50$, primary scaling leaves the all-horizon upper-certification counts unchanged in every reported synthetic condition. The allocations cover statements within each arm, not selection across arms. Fresh synthetic calibration supports those within-arm claims. Reusing the main-study streams or changing fitting seeds on the same PenDigits cohort does not supply an independent calibration sample for the additional preprocessing choices. The results indicate that the effect of preprocessing depends on both the task and the forecasting architecture. Preprocessing can be selected using validation data before calibration, as in the matched comparison below; alternatively, selecting among calibrated arms requires a failure-probability allocation that covers all such arms.

S6 A matched comparison of geometric representations

Design. To separate representation and decoder effects, we compare KAHM, RBF, and affine-hull geometry with three decoders: direct fitting under the block-norm budget in (16), direct fitting under an ordinary Frobenius-norm budget, and feature rollout. We use *head* as a short name for these decoder choices. Every geometry–head pair receives the same prefix dimensions $2, 6, 14, 30$, two history lengths, and four preprocessing options. Training, horizons, origins, regularization, and

geometry subsampling follow the original study. Equal numerical budgets and feature counts do not imply equal statistical capacity or computational cost.

Five seeds, 5701–5705, generate fresh episodes for all six synthetic tasks. PenDigits uses new fitting randomness within the same development and held-out writer cohorts. Before calibration, each geometry–head pair selects preprocessing, history, and dimension by validation MSE across horizons, with ties resolved by feature count, history, preprocessing order, and name. PenDigits selects coordinates separately, then averages their losses and horizons within each writer, giving the 13 writers equal weight. Its seed variation describes repeated fitting on that cohort.

Confidence accounting. Within each synthetic task and seed, the four preprocessing arms receive 0.01 each: 0.008 for candidate risk bounds and 0.002 for refinement comparisons. The remaining 0.01 is divided among six primary geometry comparisons. For each, the alternative's loss minus KAHM's loss is averaged within each independent calibration episode over horizons. These differences lie in $[-4, 4]$; the paired empirical Bernstein bound (31) uses range length 8 and tail probability $0.01/6$. The total event has confidence at least 95% within that synthetic task–seed combination. Fixed-dimension and head-selected analyses are descriptive. PenDigits was examined in earlier studies, so its follow-up endpoints and paired writer effects provide no new confirmatory population guarantee, even under an iid-writer assumption.

Results and their limits. All 13,440 candidate fits completed. Table S10 combines absolute MSE, its variation across fits, relative changes, and paired seed signs for all 42 contrasts. KAHM has a lower mean MSE in 28 contrasts and a higher mean MSE in 14. These contrasts share models and tasks, so their signs cannot be treated as independent observations. On nonlinear AR, KAHM has lower MSE than both geometric alternatives under every head and seed. Its block-direct MSE 0.00456 also improves on ridge AR 0.00511 and quadratic regression 0.00523.

On PenDigits, KAHM reduces MSE relative to RBF by 15.6%, 17.2%, and 17.9% across the three heads, with lower loss in all five seeds. Its reductions relative to affine geometry are 12.7%, 1.9%, and 1.4%, with four, four, and three favorable seeds. Conventional regressors remain stronger: Frobenius-direct KAHM MSE is 0.00504, compared with 0.00184 for ridge AR and 0.00102 for quadratic regression.

On Van der Pol, KAHM improves on RBF by 14.4–17.1%, but exceeds affine MSE by 9.5%, 102.7%, and 92.8% across the heads. Corresponding Duffing excesses are 13.1%, 70.6%, and 79.9%. Affine geometry has lower MSE than KAHM under Frobenius and rollout heads in every seed on both oscillators. Mean effects can also oppose most paired signs: favorable quasiperiodic block-direct means occur with lower KAHM loss in only two of five seeds.

The complete fixed-dimension curves are released as `paper/figures/matched_geometry_dimensions.pdf`, generated from all fixed-dimension records. On PenDigits, KAHM has lower mean MSE than RBF at every dimension and head; its ranking against affine geometry changes with dimension. On nonlinear AR, KAHM has lower mean MSE than both alternatives at 14 and 30 features, whereas the ranking varies at smaller dimensions. Because histories and preprocessing are reselected, these comparisons are not tests of nesting for fixed blocks.

Only one of the 180 planned synthetic improvement comparisons is certified: Van der Pol seed 5703, KAHM rollout versus RBF rollout, with lower MSE-improvement endpoint 0.00107. All others remain inconclusive. This statement concerns two fitted predictors within the declared event; it is distinct from the descriptive mean effects and implies no population comparison on PenDigits.

Task	Head	MSE $\times 10^3$: mean (SD) KAHM	RBF	Affine	KAHM change (wins) vs. RBF	vs. Affine
Harmonic	Block	49.77 (1.33)	50.97 (0.98)	52.45 (0.78)	-2.4% (5)	-5.1% (5)
	Frobenius	47.67 (1.31)	48.41 (1.13)	44.94 (1.13)	-1.5% (5)	+6.1% (0)
	Rollout	52.26 (1.04)	52.38 (0.98)	51.66 (1.42)	-0.2% (3)	+1.2% (1)
Quasiperiodic	Block	34.26 (1.32)	34.95 (0.70)	34.74 (2.82)	-2.0% (2)	-1.4% (2)
	Frobenius	33.72 (1.30)	34.30 (0.61)	25.92 (2.72)	-1.7% (3)	+30.1% (0)
	Rollout	39.22 (3.18)	41.75 (3.83)	37.95 (4.58)	-6.1% (2)	+3.3% (1)
Nonlinear AR	Block	4.56 (0.05)	4.69 (0.04)	4.77 (0.04)	-2.7% (5)	-4.4% (5)
	Frobenius	4.56 (0.05)	4.69 (0.04)	4.77 (0.04)	-2.7% (5)	-4.4% (5)
	Rollout	4.57 (0.08)	4.81 (0.07)	4.81 (0.05)	-5.0% (5)	-4.9% (5)
Logistic	Block	165.48 (4.28)	167.18 (2.74)	173.27 (2.22)	-1.0% (3)	-4.5% (5)
	Frobenius	160.66 (5.06)	161.29 (3.38)	170.63 (2.33)	-0.4% (2)	-5.8% (5)
	Rollout	167.63 (5.93)	164.90 (5.64)	177.74 (2.40)	+1.7% (1)	-5.7% (5)
Van der Pol	Block	52.92 (5.56)	61.82 (3.52)	48.35 (3.07)	-14.4% (5)	+9.5% (2)
	Frobenius	26.42 (4.55)	30.92 (7.17)	13.03 (2.92)	-14.5% (3)	+102.7% (0)
	Rollout	33.36 (4.97)	40.23 (16.47)	17.30 (3.63)	-17.1% (4)	+92.8% (0)
Duffing	Block	24.82 (1.60)	23.91 (2.21)	21.95 (2.32)	+3.8% (3)	+13.1% (1)
	Frobenius	23.41 (1.76)	21.21 (2.77)	13.72 (1.51)	+10.4% (0)	+70.6% (0)
	Rollout	27.41 (3.03)	27.02 (4.58)	15.23 (1.41)	+1.4% (3)	+79.9% (0)
PenDigits	Block	5.14 (0.32)	6.09 (0.22)	5.89 (1.15)	-15.6% (5)	-12.7% (4)
	Frobenius	5.04 (0.32)	6.09 (0.22)	5.14 (0.70)	-17.2% (5)	-1.9% (4)
	Rollout	5.46 (0.40)	6.65 (0.34)	5.54 (0.97)	-17.9% (5)	-1.4% (3)

Table S10: Matched geometry comparison after validation-only selection from identical preprocessing, history, and dimension options. Absolute all-horizon test MSE and its sample standard deviation across five fitting repetitions are scaled by 10^3. Relative changes use ratios of mean MSEs; negative values favor KAHM. Wins count repetitions with lower KAHM MSE out of five. PenDigits averages both coordinates and weights the same 13 writers equally. Every effect is descriptive; calibrated comparisons are reported separately.

S7 Speech-energy forecasting with group-level calibration

This second real-data domain tests whether a larger number of documented recording groups permits useful calibration. We use the official Mini Speech Commands subset (Warden, 2018; TensorFlow, 2026), with an eligibility rule, group split, observable, and model grid specified before downloading or inspecting the audio.

Observable and sampling units. Of 8000 source clips, 7178 have the prescribed mono, 16 kHz, signed 16-bit, exactly one-second format. They belong to 1630 filename identifiers, using the source convention before `_nohash_`. All clips from an identifier remain together. A fixed random permutation assigns 244 identifiers to training, 163 to validation, 978 to calibration, and 245 to testing. These contain 1159, 677, 4256, and 1086 clips, respectively. Identifiers denote recording sessions for web volunteers; they are not biometric confirmation of distinct people. Confidence

statements assume iid group-record bundles and independence from development data. Identifier separation supports the evaluation design but does not establish that population sampling assumption.

Each clip is divided into 100 nonoverlapping frames of 160 samples. For frame $t \in \{0, \dots, 99\}$ and sample index $j \in \{1, \dots, 160\}$, let x_{tj} be the signed pulse-code modulation (PCM) amplitude divided by 32768. The observed energy is

$$e_t = \frac{10 \log_{10}\Big(\max\Big\{160^{-1} \sum_{j=1}^{160} x_{tj}^2, 10^{-6}\Big\}\Big) + 60}{60} \in [0, 1].$$

The implementation clips numerical output to $[0, 1]$. A frame is observed at its end, so this transformation uses no future samples. It represents power in decibels relative to full scale, floored at -60 dB. Multiplying normalized RMSE by 60 gives RMSE in dB for this floored observable, not waveform error or error in unfloored acoustic power. The source selects and quality-screens one-second recordings; our complete-clip eligibility further restricts the population. This is retrospective forecasting within released clips, without per-utterance normalization by future means, peaks, or extrema.

Histories contain 4 or 12 frames, horizons are 1, 4, or 12 frames (10, 40, or 120 ms), and all candidates share origins $11, 15, \dots, 87$. Losses average over origins and clips within each identifier, then weight identifiers equally. Thousands of clips and windows do not replace the 978 independent calibration units assumed by the theorem.

Development and calibration. Three fitting seeds use the same group split. Each samples 4096 training histories with equal identifier probability and evaluates all validation groups with equal weights. The prespecified study compares 486 candidates per fit across eight families (Table S11). The geometric families share prefix sizes $2, 6, 14, 30$, standardization or raw inputs, budgets $1, 4$, and ridge parameters $10^{-4}, 10^{-2}$. RBF and affine affinity multipliers are $0.25, 1, 4$; KAHM uses its inherited reconstruction. Conventional regression heads use ridge regularization without a coefficient-norm constraint; thus their comparison with the geometric heads does not control statistical capacity. Appendix S8.6 specifies their complete grids. Every family has both histories. Selection minimizes validation MSE across horizons; all 24 selected models are frozen before held-out losses are evaluated.

For each selected model and horizon, calibration gives one bounded loss in $[0, 1]$ per identifier. The bounded-mean relative-entropy upper bound in Appendix B.3 uses tail probability $0.05/72$, covering all $3 \times 8 \times 3$ statements jointly. Prespecified tolerances of 6, 12, and 18 dB are explicit resolution levels, not claimed application requirements. Model choices and calibration formulas remain fixed during held-out evaluation. Appendix S8.6 describes the scoring order and its verification.

Findings. At 120 ms, mean test RMSE is 12.73 dB for KAHM, 12.51 for ridge AR, 12.47 for quadratic regression, and 12.39 for Gaussian-kernel regression. The latter three also outperform KAHM at shorter horizons. KAHM improves on long-horizon persistence, whose mean is 14.70 dB. These are descriptive accuracy comparisons; no separately calibrated superiority claim is made. The displayed standard deviations describe fitting variation on the same 245 test groups.

All six learned families satisfy the all-horizon 18 dB upper criterion in all three fits. Persistence and the constant model do not; no family passes the all-horizon 6 or 12 dB criterion. Thus the larger calibration cohort can resolve a declared tolerance under the stated sampling model, while leaving tighter requirements unresolved. The experiment supports the certification procedure without establishing that its tolerance is sufficient for an audio application or that KAHM is the best

Family	Test RMSE in dB: mean (SD) 10 ms	40 ms	120 ms	Maximum upper RMSE in dB 10 ms	40 ms	120 ms	All ≤ 18 dB Fits / 3
KAHM blocks	4.00 (0.15)	7.58 (0.14)	12.73 (0.06)	8.15	11.64	16.37	3/3
RBF blocks	3.79 (0.01)	7.45 (0.01)	12.72 (0.00)	7.83	11.41	16.30	3/3
Affine-hull blocks	4.28 (0.43)	7.77 (0.24)	12.81 (0.05)	8.80	11.98	16.41	3/3
Ridge AR	3.49 (0.00)	7.22 (0.01)	12.51 (0.01)	7.53	11.19	16.15	3/3
Quadratic	3.52 (0.01)	7.24 (0.03)	12.47 (0.03)	7.55	11.21	16.12	3/3
Gaussian kernel	3.51 (0.00)	7.17 (0.01)	12.39 (0.02)	7.57	11.15	16.02	3/3
Persistence	3.56 (0.00)	7.71 (0.00)	14.70 (0.00)	7.58	11.61	18.08	0/3
Constant	15.67 (0.01)	15.63 (0.01)	15.49 (0.01)	19.21	19.17	19.04	0/3

Table S11: Speech-energy forecasting on 245 held-out identifiers. Test means and sample standard deviations use three fits on the same group split. Upper columns take the maximum over those fits, with simultaneous coverage of all 72 selected family–fit–horizon statements under the stated iid group-record assumption. All six learned families pass the 18 dB requirement at every horizon in each fit; none passes the all-horizon 6 or 12 dB requirement. Errors concern the floored energy observable.

	Standard	Raw	Min–max	Clipped	$\Delta\%$	Fits
KAHM	22.570 (0.197)	21.800 (0.441)	22.251 (0.092)	22.251 (0.092)	-1.41	3/3
RBF	21.466 (0.034)	21.450 (0.023)	21.304 (0.132)	21.304 (0.132)	-0.76	3/3
Affine hull	22.493 (0.665)	22.995 (0.661)	22.547 (0.104)	22.547 (0.104)	$+0.24$	1/3

Table S12: Descriptive speech scaling comparison on the same 245 test groups. Entries are all-horizon MSE means (sample standard deviations across three fits), multiplied by 10^3. Each fit averages the three horizon MSEs with equal weight. All model choices within an arm use validation data. $\Delta\%$ compares the min–max and standardized means; negative values favor min–max. "Fits" counts the paired fits with lower min–max MSE. The clipped arm shares training extrema and clips held-out inputs to $[-1, 1]$. This reused-cohort follow-up supplies no new confidence or certification claim.

forecaster. Shared geometry costs and inference times are archived separately and imply no general computational advantage.

Input scaling on the same recording cohort. A descriptive comparison evaluates standardization, raw energy histories, training min–max scaling to $[-1, 1]$, and its clipped variant for all three geometric families. It retains the same observations, 4096 sampled training histories per seed, and decoder grids. Each transformation is fitted before the 1600-history geometry subsample. There are 2688 candidate fits in total; validation selects one model within each family, seed, and preprocessing arm. All 36 selections are frozen before test scoring. Because this analysis reuses the previously examined recording cohort, its results provide no independent confirmation or new risk certificate. Calibration groups are not scored in this comparison.

Table S12 shows that min–max scaling lowers KAHM's mean all-horizon MSE by 1.41% relative to standardization, with lower MSE in each of the three fits. Raw energy histories nevertheless give

lower KAHM MSE than either min–max variant in every fit. RBF min–max MSE decreases by 0.76% with improvement in all three fits; affine-hull mean MSE increases by 0.24%, with improvement in only one fit. Clipped and unclipped results agree at the displayed precision. The ridge AR, quadratic, and Gaussian-kernel regressors from Table S11 retain lower test MSE than every KAHM scaling arm at every horizon and seed. These comparisons describe the specified implementations and the same 245 test groups.

S8 Experimental protocols and comparison models

S8.1 Episode laws

Each row of a generated data array is one independent episode. In the following definitions, t is an integer time index, y_t is the observed scalar signal, and $\mathrm{Unif}[a, b]$ denotes the uniform distribution on $[a, b]$. All random draws are independent unless an episode-level parameter is explicitly shared across times. The observation range is contained in $[-1, 1]$ for all four main systems. Clipping forecasts to this interval gives scalar squared losses in $[0, 4]$.

1. **Harmonic.** Draw one phase $\theta \sim \mathrm{Unif}[0, 2\pi]$ and one angular frequency $\nu \sim \mathrm{Unif}[.12, .28]$ per episode. Define
$$y_t = .85\cos(\theta + \nu t) + \eta_t, \qquad \eta_t \sim \mathrm{Unif}[-.03, .03],$$
where η_t is observation noise. Thus $|y_t| \le .88$.

2. **Quasiperiodic.** Draw phases θ_1 and θ_2 independently from $\mathrm{Unif}[0, 2\pi]$, and draw $\nu \sim \mathrm{Unif}[.10, .22]$ once per episode. Set
$$y_t = .55\cos(\theta_1 + \nu t) + .30\cos(\theta_2 + \sqrt{2}\nu t) + \eta_t, \qquad \eta_t \sim \mathrm{Unif}[-.03, .03].$$
The two angular frequencies are ν and $\sqrt{2}\nu$. Again $|y_t| \le .88$.

3. **Nonlinear autoregression.** Initialize y_0 and y_1 independently from $\mathrm{Unif}[-.8, .8]$. For $t \ge 1$, apply
$$y_{t+1} = \tanh\{1.25y_t - .65y_{t-1} + .25\sin(3y_t) + \eta_{t+1}\}, \qquad \eta_{t+1} \sim \mathrm{Unif}[-.10, .10],$$
where η_{t+1} is an innovation and $\tanh$ denotes the hyperbolic tangent. The implementation discards the first 80 stored values before retaining a length-72 episode. The two initial values occupy indices zero and one; recursive updates start at index two. The hyperbolic tangent ensures the stated output bound.

4. **Logistic.** Draw the episode parameter $r \sim \mathrm{Unif}[3.65, 3.95]$ and initial internal state $u_0 \sim \mathrm{Unif}[.05, .95]$, then iterate $u_{t+1} = ru_t(1 - u_t)$. After 40 burn-in updates, store the next 72 updated values, transformed to the observed scale by $y = 2u - 1$. Because $r \le 4$, updates preserve $u \in [0, 1]$ and hence $y \in [-1, 1]$. Within-episode dynamics can be chaotic. Independence between episodes follows from independent initialization, without requiring a mixing argument.

The finite generation and origin protocol defines the risk being estimated. No assumption of stationarity after burn-in or sampling from an invariant distribution is needed. Each generator has its own observation scale and horizon interpretation; we do not average errors across generators as if they were a common physical measure.

S8.2 Oscillator protocol

For continuous time t, position $x(t)$ and velocity $v(t)$ satisfy

$$\dot{x} = v, \qquad \dot{v} = (1 - x^2)v - x \qquad \text{(Van der Pol)},$$
$$\dot{x} = v, \qquad \dot{v} = -0.2v + x - x^3 + 0.3\cos(1.2t + \varphi) \qquad \text{(Duffing)},$$

where dots denote time derivatives. Initial position and velocity are independent $\mathrm{Unif}[-2, 2]$ draws; Duffing phase $\varphi \sim \mathrm{Unif}[0, 2\pi]$ is also independent per episode. Fourth-order Runge–Kutta uses step 0.025 and discards ten time units. The 72 sensor observations are

$$y_j = 0.95\tanh(x(10 + 0.1j)) + \eta_j, \qquad \eta_j \overset{\text{iid}}{\sim} \mathrm{Unif}[-0.02, 0.02], \quad j = 0, \ldots, 71.$$

Their range $[-0.97, 0.97]$ lies within the certificate range $[-1, 1]$. A prespecified half-step check on 16 fixed initializations, with phases and noise shared between steps 0.025 and 0.0125, gives maximum sensor changes 1.69×10^{-6} (Van der Pol) and 3.49×10^{-6} (Duffing), below 10^{-3}. This checks numerical consistency without a rigorous continuous-time integration-error bound.

S8.3 PenDigits acquisition and development

The original UNIPEN files contain 7,494 development and 3,498 test digits, with nominal 100 ms sampling. Every coordinate is checked against the acquisition range $[0, 500]$; no stroke centering, alignment, or resampling is applied, and pen-up gaps are excluded. Writer identifiers include the source split; approximately 250 records per identifier support the inference described in Section S4.5.

The first 24 development writers supply training and the remaining six validation. Each of three seeds selects 20 eligible strokes per training writer (480 total); validation uses all 717 eligible strokes. Histories have lengths $m \in \{4, 8\}$, common zero-based origins $7, 11, 15, 19, 23, 27$, and horizons $1, 2, 4$. The 40-candidate family follows the main comparison design, with successive block sizes $2, 4, 8, 16$, $\kappa = 4$, ridge $\lambda = 10^{-4}$, and affinity exponent $\omega = 16$. Validation weights writers equally and averages MSE over horizons. Normalized RMSE 0.01 equals 2.5 acquisition pixels. For the additional iid-writer calculation, two coordinates, three fits, 40 candidates, and three horizons give 720 one-sided statements. With 13 writers, the Hoeffding addition is $4\sqrt{\log(720/0.05)/(2 \cdot 13)} = 2.42741$.

S8.4 Random streams and configuration

Main fitting seeds are $1701, \ldots, 1705$. For seed s, separate NumPy streams use $100s+11$, $100s+23$, $100s + 47$, and $100s + 89$ for training, validation, calibration, and testing. Clustering and retained blocks receive deterministic offsets from the training seed and history length. Archived configurations specify these offsets and random-generator versions. Geometry uses at most 1600 training rows; decoders use all 1664 rows (128 episodes, 13 common origins). Overlapping training rows are permitted. Validation and test losses weight episodes equally and origins equally within episodes; selection averages MSE over the three horizons.

S8.5 Comparison definitions

In the main synthetic campaigns, RBF and affine-hull comparators use the same type of k-means partition, with five initializations and at most 150 iterations. Here k is the number of coordinates

in the block. To define their scores, suppress the block index. Let $\xi \in \mathbb{R}^{d_x}$ be a preprocessed history, where d_x is its input dimension. For cluster $c \in \{1, \dots, k\}$, let $\mu_c \in \mathbb{R}^{d_x}$ be its fitted center, $\mathcal{I}_c$ the nonempty set of assigned geometry-training indices, and $n_c = |\mathcal{I}_c|$. Define the cluster's squared-radius scale and the global RBF scale by

$$s_c = \max\left\{\frac{1}{n_c}\sum_{i\in\mathcal{I}_c}\|\xi_i - \mu_c\|_2^2,\ 10^{-6}\right\}, \qquad \beta = \mathrm{median}_{1\le c\le k}\, s_c,$$

where median is the usual sample median. The RBF score is

$$a_c^{\mathrm{RBF}}(\xi) = \frac{\|\xi - \mu_c\|_2^2}{2\beta}.$$

Thus every cluster uses the same bandwidth, while the positive floor prevents division by a zero radius.

For the affine-hull comparator, take the singular value decomposition of the matrix whose rows are $(\xi_i - \mu_c)^\top$, $i \in \mathcal{I}_c$. When $n_c \ge 2$, let V_c contain its first $k_c = \min\{2, d_x - 1, n_c\}$ right singular vectors as columns. For a singleton cluster, set $k_c = 0$ and use an empty basis. Then $P_c = V_c V_c^\top$ is the orthogonal projection onto the selected local subspace; $P_c = 0$ for an empty basis. The constraint $k_c \le d_x - 1$ leaves an orthogonal direction whenever one is available. Define

$$r_c = \max\left\{\frac{1}{n_c}\sum_{i\in\mathcal{I}_c}\|(I_{d_x} - P_c)(\xi_i - \mu_c)\|_2^2,\ 0.05 s_c,\ 10^{-6}\right\},$$
$$a_c^{\mathrm{AH}}(\xi) = \frac{\|(I_{d_x} - P_c)(\xi - \mu_c)\|_2^2}{r_c} + 0.1\frac{\|\xi - \mu_c\|_2^2}{s_c},$$

where I_{d_x} is the identity matrix. The first term measures distance from the local affine subspace; the second penalizes distance from its center. The floors in r_c keep the normalization positive even when the training residuals vanish.

For either choice $a_c = a_c^{\mathrm{RBF}}$ or $a_c = a_c^{\mathrm{AH}}$, the block coordinates are

$$\Phi_c(\xi) = \frac{\exp(-a_c(\xi))}{\sum_{j=1}^k \exp(-a_j(\xi))}.$$

Normalization is evaluated in log space. The score multiplier is one in the original synthetic campaigns; subsequent studies specify their expanded grids. Both alternatives use the retained-block decoder.

Hard KAHM states replace the final block by a one-hot vector: its largest coordinate becomes one and all others become zero. Ridge AR uses the original causal history. Quadratic regression uses training-standardized histories and all monomials of degree one or two, including cross-products; it does not add a constant feature because the intercept is unpenalized. Both direct decoders use the stated Frobenius constraint and ridge coefficient. Persistence repeats the latest observation at every horizon. The constant model predicts the training target mean separately for each horizon and has sensitivity budget zero.

For the iterated baselines, let $z_t \in \mathbb{R}^{1\times p}$ be the row vector of p features at time t. Ridge regression fits an affine feature transition and a scalar observation decoder,

$$z_{t+1} \approx z_t M + b, \qquad y_t \approx z_t v + c,$$

where $M \in \mathbb{R}^{p\times p}$ is the transition matrix, $b \in \mathbb{R}^{1\times p}$ is its intercept, $v \in \mathbb{R}^p$ is the observation coefficient vector, and $c \in \mathbb{R}$ is the observation intercept. For a positive integer horizon h, composing these fitted maps gives the un-clipped forecast

$$\widetilde{y}_{t+h} = z_t M^h v + b\left(\sum_{\ell=0}^{h-1} M^\ell\right) v + c,$$

with M^0 the identity matrix. Only this final scalar forecast is clipped. The effective output coefficient is $M^h v$; its Euclidean norm is the frozen Lipschitz budget relative to Euclidean feature distance at horizon h. The certificate evaluates this fitted forecast directly and requires no spectral-stability assumption, exact feature closure, or approximation to the trajectory distribution.

S8.6 Speech model grid and evaluation order

The prespecified speech study uses fitting seeds $8601, 8602, 8603$ and history lengths $m \in \{4, 12\}$. Each of the KAHM, RBF, and affine-hull families uses the four retained prefixes, two preprocessing choices (standardization and raw inputs), budgets $\kappa \in \{1, 4\}$, and ridge coefficients $\lambda \in \{10^{-4}, 10^{-2}\}$. For RBF and affine-hull features, an additional multiplier $\rho \in \{0.25, 1, 4\}$ changes the association weights to values proportional to $\exp(-\rho a_c(\xi))$, with a_c defined above. KAHM retains affinity exponent 16, at most 1600 geometry-training histories, and at most 24 reference points per local model. The resulting candidate counts are 64 for KAHM and 192 for each geometric comparator.

Linear and quadratic regression use training-standardized histories, with ridge coefficients λ in $\{10^{-6}, 10^{-4}, 10^{-2}, 10^{-1}\}$. Gaussian-kernel regression uses the same standardization and a 128-landmark Nyström approximation to $k(z, z') = \exp(-\gamma\|z - z'\|_2^2)$. Its kernel parameter is $\gamma = r/d_{mmed}$, where $r \in \{0.25, 1, 4\}$ and d_{mmed} is the median positive squared distance in a seeded subsample of 512 training rows; the implementation sets $d_{mmed} = 1$ if all those distances vanish. The kernel ridge grid is $\{10^{-4}, 10^{-2}, 10^{-1}\}$. All three regression heads have an unpenalized intercept and no coefficient-norm constraint. Together with two history-specific instances each of persistence and the constant predictor, these grids give 486 candidates per fit. All predictions are clipped to the fixed target interval $[0, 1]$.

Every model and selection rule is fixed before held-out scoring. The first execution evaluated calibration and test losses consecutively for each model, whereas the prescribed order completes all calibration calculations before test scoring. Repeating the calculations in the prescribed order gives identical selected predictors, group losses, and confidence endpoints; no choice depends on the intervening test losses. The two executions therefore represent one study. The source versions, evaluation-order records, and independent reconstruction are retained in the repository.

S8.7 Reproduction and study provenance

The repository provides configurations, source hashes, candidate definitions, per-episode scores, and scripts for independently reconstructing the reported confidence endpoints. Episode-score archives support reconstruction; fixed-seed runners regenerate large files omitted from the repository. Archived environments specify numerical-library versions; bitwise agreement across architectures is not assumed. Complete commands for all studies, asset generation, and manuscript compilation are given in `extensions/certified_forecasting/docs/REPRODUCIBILITY.md`. Run-

ners preserve failed attempts and refuse to overwrite completed runs. Original-seed and confirmation input-scaling analyses remain separate, with the scope in Section S5.

S8.8 Matched-geometry protocol

Each preprocessing arm has 72 matched scalar candidates and 12 controls. The refinement allocation covers 54 adjacent-prefix horizon comparisons: three geometries, two histories, three adjacent prefix pairs, and three horizons. The complete schedule and configuration hashes precede fitting; Section S6 specifies selection and joint confidence allocation. The result archive contains all 13,440 candidates, paired seed effects, horizon-specific results, and PenDigits writer comparisons. Table S10 combines absolute errors, fitting variation, and relative effects. The asset-generation scripts read these complete analysis records without refitting models or selecting configurations. Each study has a separate configuration and provenance record.

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